ICSE Class X Mathematics

Competency-Focused Practice Questions (MCQ)

A retailer buys an article at its listed price from a wholesaler and sells it to a consumer in the same state after marking up the price by 20%. List price Rs. 2500, GST rate 12%. What is the tax liability of the retailer to the central government?

(a) Rs. 0
(b) Rs. 15
(c) Rs. 30
(d) Rs. 60

Answer to Q1

Answer: (c) Rs. 30

Input GST $= 12\%$ of $2500 = 300$. Selling price $= 3000$, output GST $= 360$. Liability $= 360 - 300 = 60$; intra-state, so split equally: CGST $= Rs.\,30$.

Dev bought an electrical fan with marked price Rs. 800. If the GST on the goods is 7%, then the SGST is:

(a) Rs. 24
(b) Rs. 28
(c) Rs. 56
(d) Rs. 80

Answer to Q2

Answer: (b) Rs. 28

GST $= 7\%$ of $800 = 56$. SGST $= \frac{1}{2} \times 56 = Rs.\, 28$.

Rs. P is deposited for n months in a recurring deposit account paying interest at $r\%$ per annum. The nature and time of interest calculated is:

(a) compound interest for n months.
(b) simple interest for n months.
(c) compound interest for one month.
(d) simple interest for one month.

Answer to Q3

Answer: (d) simple interest for one month

RD interest is simple interest: $I = P \times \frac{n(n+1)}{2} \times \frac{r}{12 \times 100}$. The sum $P \times \frac{n(n+1)}{2}$ is the equivalent principal for *one month*.

Anwesha planned an RD of Rs. 1000 per month for 1 year at 5% p.a. simple interest. The rate fell to 4%. How much must she deposit monthly for 1 year so her interest stays the same?

(a) Rs. 12325
(b) Rs. 1250
(c) Rs. 1200
(d) Rs. 1000

Answer to Q4

Answer: (b) Rs. 1250

Original $I = 1000 \times \frac{12 \cdot 13}{24} \times \frac{5}{100} = 325$. New: $P \times 6.5 \times 0.04 = 325 \Rightarrow P = Rs.\,1250$.

Mr. Das invests in Rs. 100, 12% shares of Company A available at Rs. 60 each. Mr. Singh invests in Rs. 50, 16% shares of Company B available at Rs. 40 each. Which statement is true?

(a) Rate of return for Mr. Das is 12%
(b) Rate of return for Mr. Singh is 10%
(c) Both have the same rate of return of 10%
(d) Both have the same rate of return of 20%

Answer to Q5

Answer: (d) Both have 20%

Das: dividend $= 12$ per share, return $= \frac{12}{60} \times 100 = 20\%$. Singh: dividend $= 16\%$ of $50 = 8$, return $= \frac{8}{40} \times 100 = 20\%$.

Amit invested in Rs. 100 shares paying a 7.5% dividend. His rate of return is 10%. The money invested to purchase 10 shares is:

(a) Rs. 250
(b) Rs. 750
(c) Rs. 900
(d) Rs. 1100

Answer to Q6

Answer: (b) Rs. 750

Dividend per share $= 7.5$. Return $= \frac{7.5}{MV} \times 100 = 10 \Rightarrow MV = 75$. For 10 shares: $10 \times 75 = Rs.\,750$.

If $-3 \leq -4x + 5$ and $x \in W$, then the solution set is:

(a) $\{\cdots -3, -2, -1, 0, 1, 2, 3 \ldots\}$
(b) $\{1, 2\}$
(c) $\{0, 1, 2\}$
(d) $\{2, 3, 4, 5\}$

Answer to Q7

Answer: (c) $\{0, 1, 2\}$

$-8 \leq -4x \Rightarrow x \leq 2$. With x a whole number: $\{0, 1, 2\}$.

If $-4x > 8y$, then

(a) $x > 2y$
(b) $x > -2y$
(c) $x < -2y$
(d) $x < 2y$

Answer to Q8

Answer: (c) $x < -2y$

Dividing by -4 reverses the inequality: $x < -2y$.

The value(s) of k for which $2x^2 - kx + k = 0$ has equal roots is/are:

(a) 0 only
(b) 4, 0
(c) 8 only
(d) 0, 8

Answer to Q9

Answer: (d) 0, 8

$D = 0$: $k^2 - 8k = 0 \Rightarrow k(k-8) = 0 \Rightarrow k = 0$ or 8.

If $x = -2$ is a solution of $x^2 + 3a - x = 0$, then the value of a is:

(a) -8
(b) -2
(c) $-1/3$
(d) $1/3$

Answer to Q10

Answer: (b) -2

$4 + 3a + 2 = 0 \Rightarrow 3a = -6 \Rightarrow a = -2.$

In solving a quadratic equation, one value of x is 233.356. The solution rounded to two significant figures is:

(a) 233.36
(b) 233.35
(c) 233.3
(d) 230

Answer to Q11

Answer: (d) 230

Two significant figures: 2 and 3; the next digit is 3, so round down and fill with zero: 230.

Right triangle ABC (right angle at B), $AB = x$ cm, $BC = y$ cm, $x - y = 7$ cm and area $= 30$ cm^2. The length of AC is:

(a) 10 cm
(b) 12 cm
(c) 13 cm
(d) 15 cm

Answer to Q12

Answer: (c) 13 cm

$\frac{1}{2}xy = 30 \Rightarrow xy = 60$. With $x - y = 7$: $x = 12$, $y = 5$. $AC = \sqrt{144 + 25} = 13$ cm.

If p, q, r are in continued proportion, then:

(a) $p : q = p : r$
(b) $q : r = p^2 : q^2$
(c) $p : q^2 = r : p^2$
(d) $p : r = p^2 : q^2$

Answer to Q13

Answer: (d) $p : r = p^2 : q^2$

$\frac{p}{q} = \frac{q}{r} \Rightarrow q^2 = pr$. Then $\frac{p}{r} = \frac{p^2}{pr} = \frac{p^2}{q^2}$.

The ratio of diameter to height of a Borosil cylindrical glass is $3 : 5$. If the actual diameter is 6 cm, the curved surface area is:

(a) 120π
(b) 60π
(c) 30π
(d) 18π

Answer to Q14

Answer: (b) 60π

$d = 6 \Rightarrow r = 3,\ h = 10.\ \text{CSA} = 2\pi rh = 2\pi(3)(10) = 60\pi.$

The polynomial $2x^3 + 3x^2 - 2x - 3$ is completely divisible by $(2x + a)$ and the quotient is $(x^2 - 1)$. One value of a is:

(a) -3
(b) -1
(c) 1
(d) 3

Answer to Q15

Answer: (d) 3

$(x^2 - 1)(2x + 3) = 2x^3 + 3x^2 - 2x - 3$. So $a = 3$.

A polynomial is $x^3 + 5x^2 - kx - 24$. Which is a factor if $k = 2$?

(a) $(x + 2)$
(b) $(x - 3)$
(c) $(x + 4)$
(d) $(x - 4)$

Answer to Q16

Answer: (c) $(x+4)$

$f(x) = x^3 + 5x^2 - 2x - 24$. $f(-4) = -64 + 80 + 8 - 24 = 0$, so $(x+4)$ is a factor.

If $A = [a \quad b]$ and $B = \begin{bmatrix} c \\ d \end{bmatrix}$, then:

(a) only AB is possible
(b) only BA is possible
(c) both AB and BA are possible
(d) both are possible and $AB = BA$

Answer to Q17

Answer: (c) both AB and BA are possible

A is 1×2, B is 2×1. AB is 1×1, BA is 2×2; different orders, so $AB \neq BA$.

Matrix $A = \begin{bmatrix} 6 & 9 \\ -4 & k \end{bmatrix}$ with $A^2 = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$. Then k is:

(a) 6
(b) -6
(c) 36
(d) ± 6

Answer to Q18

Answer: (b) -6

$A^2 = \begin{bmatrix} 0 & 54+9k \\ -24-4k & k^2-36 \end{bmatrix}$. $54 + 9k = 0 \Rightarrow k = -6$, which also satisfies the other entries.

If the sum of n terms of an AP is $S_n = n^2 - n$, the third term is:

(a) 2
(b) 4
(c) 6
(d) 9

Answer to Q19

Answer: (b) 4

$a_3 = S_3 - S_2 = 6 - 2 = 4.$

Which of the following is NOT a geometric progression?

(a) $1/3, 1, 3, 9$
(b) $1/5, 1/5, 1/5, 1/5$
(c) $-2, 4, -8, 16$
(d) $2, 0, 4, 0, 8, 0$

Answer to Q20

Answer: (d) $2, 0, 4, 0, 8, 0$

(a) $r = 3$, (b) $r = 1$, (c) $r = -2$. In (d), ratios $0/2 = 0$ and $4/0$ is undefined; not a GP.

G(5,−5) is the centroid of $\triangle ABC$ with A(3,−3), B(2,−6), C(x, y). D lies on BC, with AD the median through G. Coordinates of D:

(a) (2,−6)
(b) (3,−6)
(c) (6,−6)
(d) (10,−6)

Answer to Q21

Answer: (c) $(6,-6)$

$\frac{3+2+x}{3} = 5 \Rightarrow x = 10$; $\frac{-3-6+y}{3} = -5 \Rightarrow y = -6$. D is midpoint of BC: $\left(\frac{2+10}{2}, \frac{-6-6}{2}\right) = (6,-6)$.

O is the origin, A is $(4, 0)$, B is $(0, 2)$, and P is the midpoint of AB (from the figure). The equation of OP is:

(a) $y = x$
(b) $2y = x$
(c) $y = 2x$
(d) $y = -x$

Answer to Q22

Answer: (b) $2y = x$

$P = (2, 1)$. Slope of OP $= \frac{1}{2}$, passing through origin: $y = \frac{x}{2} \Rightarrow 2y = x$.

Line l_1 is parallel to l_2, and makes 45° with the x-axis. Line l_3 is perpendicular to l_1. Slopes of l_2 and l_3 respectively are:

(a) 1, 1
(b) $-1, -1$
(c) 1, -1
(d) -1, 1

Answer to Q23

Answer: (c) 1, -1

$m_1 = \tan 45^\circ = 1$; parallel: $m_2 = 1$. Perpendicular: $m_3 = -1/m_1 = -1$.

Which line cuts the positive x-axis and positive y-axis at equal distances from the origin?

(a) $3x + 3y = 6$
(b) $5x + 10y = 10$
(c) $-x + y = 1$
(d) $10x + 5y = 5$

Answer to Q24

Answer: (a) $3x + 3y = 6$

$3x + 3y = 6 \Rightarrow \frac{x}{2} + \frac{y}{2} = 1$: both intercepts are 2. Others: (b) 2 and 1; (c) -1 and 1; (d) $\frac{1}{2}$ and 1.

Stations A, B, C, P, Q joined by straight tracks; $PQ \parallel BC$, $PQ = 20$ km, $BC = 50$ km, $PB = 18$ km (P on AB). Time for a train at 90 km/hr to go from A to B by the shortest route:

(a) 8 minutes
(b) 12 minutes
(c) 16.8 minutes
(d) 20 minutes

Answer to Q25

Answer: (d) 20 minutes

$\triangle APQ \sim \triangle ABC$: $\frac{AP}{AB} = \frac{20}{50} = \frac{2}{5}$. $\frac{AP}{AP+18} = \frac{2}{5} \Rightarrow AP = 12$, $AB = 30$ km. Time $= \frac{30}{90}$ h $= 20$ min.

In $\triangle ABC$ and $\triangle DEF$ (not to scale), $\angle C = \angle F$ and $\frac{AB}{DE} = \frac{BC}{EF}$. Then:

(a) $\triangle ABC \sim \triangle DEF$

(b) $\triangle BCA \sim \triangle DEF$

(c) $\triangle CAB \sim \triangle DEF$

(d) similarity cannot be determined

Answer to Q26

Answer: (d) cannot be determined

The equal angle C is not the angle *included* between sides AB and BC (it's between BC and CA). This is SSA, which doesn't prove similarity.

ST is not parallel to PQ (T on RQ, S on RP). The necessary and sufficient condition for $\triangle PQR \sim \triangle TSR$ is:

(a) $\angle PQR = \angle STR$
(b) $\angle QPR = \angle TSR$
(c) $\angle PQR = \angle TSR$
(d) $\angle PRQ = \angle RST$

Answer to Q27

Answer: (c) $\angle PQR = \angle TSR$

$\angle R$ is common. For $P \leftrightarrow T,\ Q \leftrightarrow S,\ R \leftrightarrow R$ we need $\angle PQR = \angle TSR$ (AA similarity).

The scale factor of a picture to Sonia's actual height is 20 cm : 1.6 m. If her height in the picture is 18 cm, her actual height is:

(a) 14.4 m
(b) 2.25 m
(c) 1.78 m
(d) 1.44 m

Answer to Q28

Answer: (d) 1.44 m

$\frac{18}{20} \times 1.6 = 1.44$ m.

O is the centre of the circle; a semicircle is drawn on OA as diameter and meets the circle at Q. $\angle APQ = 20^\circ$. The measure of $\angle OAQ$ is:

(a) 25°
(b) 40°
(c) 50°
(d) 65°

Answer to Q29

Answer: (c) 50°

$\angle AOQ = 2\angle APQ = 40^\circ$ (angle at centre). $\angle OQA = 90^\circ$ (angle in a semicircle). So $\angle OAQ = 180^\circ - 90^\circ - 40^\circ = 50^\circ$.

O is the centre; PQ is a tangent at A; $\angle ABC = 50^\circ$. Values of x ($\angle QAC$), y ($\angle AOC$), z ($\angle OCA$) respectively are:

(a) $50^\circ, 100^\circ, 40^\circ$
(b) $50^\circ, 50^\circ, 65^\circ$
(c) $40^\circ, 80^\circ, 50^\circ$
(d) $50^\circ, 25^\circ, 78^\circ$

Answer to Q30

Answer: (a) $50^\circ, 100^\circ, 40^\circ$

Alternate segment theorem: $x = \angle ABC = 50^\circ$. Angle at centre: $y = 2 \times 50^\circ = 100^\circ$. $OA = OC$: $z = \frac{180^\circ - 100^\circ}{2} = 40^\circ$.

PT and QT are tangents to a circle with $\angle TPS = 45^\circ$, $\angle TQS = 30^\circ$. The value of x ($\angle PSQ$) is:

(a) 30°
(b) 45°
(c) 75°
(d) 105°

Answer to Q31

Answer: (d) 105°

Alternate segment theorem: $\angle PQS = \angle TPS = 45^\circ$ and $\angle QPS = \angle TQS = 30^\circ$. In $\triangle PSQ$: $x = 180^\circ - 45^\circ - 30^\circ = 105^\circ$.

A cylindrical metallic wire is stretched to double its length. Which will NOT change?

(a) Its curved surface area
(b) Its total surface area
(c) Its volume
(d) Its radius

Answer to Q32

Answer: (c) Its volume

Stretching reshapes the metal without adding or removing any; volume is conserved. The length doubles, so the radius decreases by $\sqrt{2}$ and the surface areas change.

A right circular cone has base radius equal to its height. If the volume is 9702 cm^3, the diameter of the base is: (use $\pi = \frac{22}{7}$)

(a) 21 cm
(b) 42 cm
(c) $21\sqrt{7}$ cm
(d) $2\sqrt{7}$ cm

Answer to Q33

Answer: (b) 42 cm

$\frac{1}{3}\pi r^3 = 9702 \Rightarrow r^3 = \frac{9702 \times 3 \times 7}{22} = 9261 \Rightarrow r = 21$. Diameter $= 42$ cm.

A solid sphere of radius 4 cm is cut into 4 identical pieces by two mutually perpendicular planes through its centre. Total surface area of one quarter piece:

(a) 24π
(b) 32π
(c) 48π
(d) 64π

Answer to Q34

Answer: (b) 32π

Curved part $= \frac{1}{4}(4\pi r^2) = 16\pi$. Two flat semicircular faces $= 2 \times \frac{1}{2}\pi r^2 = 16\pi$. Total $= 32\pi$.

Two identical solid hemispheres are joined to form a sphere. Ratio of the total surface areas of the two hemispheres to the surface area of the sphere:

(a) 1 : 1
(b) 3 : 2
(c) 2 : 3
(d) 2 : 1

Answer to Q35

Answer: (b) 3 : 2

Each hemisphere: $3\pi r^2$; two: $6\pi r^2$. Sphere: $4\pi r^2$. Ratio $= 6 : 4 = 3 : 2$.

$\text{cosec}^2\,\theta + \sec^2\theta$ is equal to:

(a) $\tan^2\theta + \cot^2\theta$
(b) $\cot\theta + \tan\theta$
(c) $(\cot\theta + \tan\theta)^2$
(d) 1

Answer to Q36

Answer: (c) $(\cot\theta + \tan\theta)^2$

$= 1 + \cot^2\theta + 1 + \tan^2\theta = \cot^2\theta + \tan^2\theta + 2\cot\theta\tan\theta = (\cot\theta + \tan\theta)^2.$

Given $a = 3\sec^2\theta$ and $b = 3\tan^2\theta - 2$, the value of $(a - b)$ is:

(a) 1
(b) 2
(c) 3
(d) 5

Answer to Q37

Answer: (d) 5

$a - b = 3(\sec^2\theta - \tan^2\theta) + 2 = 3(1) + 2 = 5.$

At a certain time, the ratio of the height of a pole to the length of its shadow is $1 : \sqrt{3}$. The angle of elevation of the sun is:

(a) 30°
(b) 45°
(c) 60°
(d) 90°

Answer to Q38

Answer: (a) 30°

$\tan\theta = \frac{1}{\sqrt{3}} \Rightarrow \theta = 30^\circ$.

A man on a ship approaching the port, towards the lighthouse, observes the top of the lighthouse. In 10 minutes, the angle of elevation changes from α to β. Then:

(a) $\alpha > \beta$
(b) $\alpha < \beta$
(c) $\alpha = \beta$
(d) $\alpha \leq \beta$

Answer to Q39

Answer: (b) $\alpha < \beta$

As the ship moves closer, the horizontal distance decreases, so $\tan\theta = \frac{h}{d}$ increases. Hence $\beta > \alpha$.

Assertion (A): The difference in class marks of the modal class and the median class of the table is 0.

Class	20–30	30–40	40–50	50–60	60–70
Freq.	1	3	2	6	4

Reason (R): Modal class and median class are always the same for a given frequency distribution.

(a) Both A and R correct, R explains A
(b) Both A and R correct, R does not explain A
(c) A is true, but R is false
(d) Both A and R are true

Answer to Q40

Answer: (c) A true, R false

Modal class = 50–60 (freq 6). $N = 16$; cumulative frequencies $1, 4, 6, 12$, so the 8th item lies in 50–60. Same class, so A is true. R is false: they can differ in general.

Assertion (A): For 11 arrayed data, the median is the middle number.
Reason (R): For the data 5, 9, 7, 13, 10, 11, 10 the median is 13.

(a) Both A and R correct, R explains A
(b) Both A and R correct, R does not explain A
(c) A is true, but R is false
(d) Both A and R are true

Answer to Q41

Answer: (c) A true, R false

With 11 values (odd), the median is the 6th value: A is true. Arranged: $5, 7, 9, 10, 10, 11, 13$; the median is 10, not 13. R is false.

Ankit can invest in company A (7%, Rs. 100 shares at Rs. 120) or company B (8%, Rs. 1000 shares at Rs. 1620).

A: Investment in Company A is better than Company B.

R: The rate of income in Company A is better than in Company B.

(a) Both true, R is the correct explanation
(b) Both true, R is not the correct explanation
(c) A is false, R is true
(d) Both A and R are false

Answer to Q42

Answer: (a)

A: $\frac{7}{120} \times 100 \approx 5.83\%$. B: $\frac{80}{1620} \times 100 \approx 4.94\%$. Income rate in A is higher, which is exactly why A is the better investment.

A: $x^3 + 2x^2 - x - 2$ is a polynomial of degree 3.
R: $x + 2$ is a factor of the polynomial.

(a) Both correct, R explains A
(b) Both correct, R does not explain A
(c) A is true, R is false
(d) Both A and R are true

Answer to Q43

Answer: (b)

Highest power is 3, so A is true. $x^2(x+2) - 1(x+2) = (x+2)(x^2-1)$, so R is true. Being a factor has no bearing on the degree, so R doesn't explain A.

A: The point $(-2, 8)$ is invariant under reflection in the line $x = -2$.
R: If a point has its x-coordinate 0, it is invariant under reflection in both axes.

(a) Both correct, R explains A
(b) Both correct, R does not explain A
(c) A is true, R is false
(d) Both A and R are true

Answer to Q44

Answer: (c) A true, R false

$(-2, 8)$ lies on $x = -2$, so it is its own image: A is true. A point $(0, y)$ is invariant in the y-axis but in the x-axis maps to $(0, -y)$ unless $y = 0$: R is false.

A die is cast with faces numbered 1 to 6 (as in the net shown). The ratio of the probability of a composite number to that of a prime number is:

(a) 2 : 3
(b) 3 : 2
(c) 1 : 3
(d) 1 : 2

Answer to Q45

Answer: (a) 2 : 3

Composite: $\{4, 6\}$, so $P = \frac{2}{6}$. Prime: $\{2, 3, 5\}$, so $P = \frac{3}{6}$. Ratio $= 2 : 3$.

$A = \begin{bmatrix} 1 & -2 \\ -3 & 4 \end{bmatrix}$, $AM = B$ where $B = \begin{bmatrix} 2 \\ 24 \end{bmatrix}$. The order of matrix M is:

(a) 2×2
(b) 2×1
(c) 1×2
(d) 4×1

Answer to Q46

Answer: (b) 2×1

A is 2×2 and AM is 2×1. Columns of $A = 2 =$ rows of M; columns of $M = 1$. So M is 2×1.

$a_1, a_2, a_3, \ldots$ and $b_1, b_2, b_3, \ldots$ are real numbers such that $a_1 - b_1 = a_2 - b_2 = a_3 - b_3 = \cdots$. Then $a_1 - b_1,\ a_2 - b_2,\ a_3 - b_3, \ldots$ forms a _____ progression.

(a) Geometric $(r = 1)$
(b) Arithmetic $(d = 1)$
(c) Geometric $(r < 1)$
(d) Arithmetic $(d = 0)$

Answer to Q47

Answer: (d) Arithmetic ($d = 0$)

All terms are equal, so the common difference is 0. It is always an AP; it would be a GP only if the constant were non-zero, so (d) is the safe choice.

The locus of a moving point that keeps a fixed distance from a fixed point is a _____.

(a) Circle
(b) Line
(c) Angle
(d) Line segment

Answer to Q48

Answer: (a) Circle

All points at a fixed distance (the radius) from a fixed point (the centre) form a circle.

The point of concurrence of the angle bisectors of a triangle is called the _____ of the triangle.

(a) centroid
(b) incentre
(c) circumcentre
(d) orthocentre

Answer to Q49

Answer: (b) incentre

The angle bisectors meet at the incentre, the centre of the inscribed circle (equidistant from all three sides).

Answer Key

1 c	2 b	3 d	4 b	5 d	6 b	7 c	8 c
9 d	10 b	11 d	12 c	13 d	14 b	15 d	16 c
17 c	18 b	19 b	20 d	21 c	22 b	23 c	24 a
25 d	26 d	27 c	28 d	29 c	30 a	31 d	32 c
33 b	34 b	35 b	36 c	37 d	38 a	39 b	40 c
41 c	42 a	43 b	44 c	45 a	46 b	47 d	48 a
49 b							